

LED lights can be as simple as a matrix of LEDs with resistor, or high-power multipurpose LEDs that are strong and long-lasting. These are designed to operate non-stop for long durations because of extremely low power consumption and very little heat dissipation. They are ideal to be used as ambient and architectural lights as well as medical and convenient lights.

The 'cold' generation of LED light leads to high efficacy because most of the energy radiates within the visible spectrum.

According to the research statistics, the response time of a person viewing an LED brakelight is quicker than brake lights powered by incandescent lights, and it can add 4 to 6 metres of safe distance while you are driving through the superhighway. Many domestic and international automobile manufacturers are already using LED-powered brake lights, appearance lights and floodlights.

How to design an LED module?

The 'LED Street Light Design and Development' article by Tsung-Chieh, Chen, from System Integration Development Division of Alliance Optotek Corp. (AOP), Taiwan, describes the various design considerations for LED modules nicely. Extracts from this article are reproduced below:

"Since there exist obvious differences between LED and traditional lamps, the thermal, optical, power mechanism and control become very important as one designs LED lighting products.

"In the past two years, LED street lights have been widely discussed since there exists a very large market and the LED's power saving performance is obvious especially in replacing the 250W or 400W mercury lamp. To build an LED streetlight, Alliance Optotek Corp. of Taiwan has analysed the key issues to meet the specifications such as guarantee of five years' LED life, light pattern to meet the roadway regulation specification, dust and water-resistant specifications, etc. Based on these

qualitative requirements, to design the LED streetlight, one needs to consider the thermal, optical and power mechanism design strategies.

"It is known that the LED life depends on the LED junction temperature. Cree, another major manufacturer of LEDs, has announced data to show the relationship of life versus maintenance of different junction temperatures. Therefore, to guarantee five years or 50,000 hours of life, the LED junction temperature has to be controlled under 75°C.

"As for the illumination uniformity of roadway, use of the batwing light pattern design of AOP is inevitable. The cut-off light pattern for glare limitation is also required. The other important subject is both the driver (power) design and the LED light module design: We know that each LED has a little different drive voltage and we can not guarantee the same drive voltage for all of the LEDs that have been built into the streetlight. So a worse circuit design will induce a non-uniform current to drive the LED, which will impact the LED's life.

"AOP knows all of the design issues that will directly affect the quality of the LED streetlight product, hence it combined thermal, optical, power and mechanism to do an overall design that ensures streetlight quality.

"So an excellent LED lighting product has to consider: thermal, optical and power mechanism (industry design).

Thermal design. As it is known, the higher the LED junction temperature, the shorter the LED life-time. To guarantee 50,000 hours of life-time, AOP has designed a streetlight in which the LED junction temperature is controlled under 75°C and heatsink temperature variation is constrained below 2°C to ensure that all of the LEDs applied on the streetlight reach the same life. They found a reasonable heatsink size to reach the design target, LED junction below 75°C and heatsink temperature variation under 2°C.

LED street light optical pattern design. As mentioned above, if the street-

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